

# Artificial intelligence in business curriculum: The pedagogy and learning outcomes

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## ABSTRACT

Artificial intelligence is rapidly reshaping the technological landscape, becoming an important component of work processes and decision-making in many domains. However, AI education has yet to catch up with the challenge of introducing this complex and very important area of technology to audiences beyond the students of computing and engineering disciplines. This is not surprising, given the lack of materials for and experience in teaching AI for non-technical audiences. In this paper we present the challenges in developing an AI course in business schools and propose a curriculum for a graduate-level business course in AI, which balances AI fundamentals with the latest developments in the field. We discuss our choices and present the results of a study, examining students' experiences, learning outcomes, and perceptions regarding the course. We offer practical guidelines for the curriculum design that other educators may adopt in designing their AI training courses.

## 1. Introduction

Thanks to big data, escalated computer hardware speed, and innovations in machine learning algorithms, research in artificial intelligence (AI) has made impressive progress in recent years. AI technologies are being developed, deployed, and used in a growing number of domains to perform complex tasks, such as car driving and speech recognition, which could have only been achieved by human beings before. AI could potentially become the most disruptive technology in human history and will have profound impacts on every aspect of our lives, especially on the labor market (Minocha et al., 2018). A recent study by McKinsey Global Institute estimates that about 70 percent of companies will have adopted at least one type of AI technology by 2030, and that 60 percent of current occupations can be automated in the next 10 years (Bughin et al., 2018). Companies and organizations will need employees who understand AI concepts and methods, are capable of using their knowledge and skills to manage AI-augmented workplace, and are well versed in collaborating with AI agents (Ransbotham, Kiron, & Gerbert, 2017). Recognizing the critical role that AI will play in defining the future international competition, many countries now regard AI as a national priority. The United States launched the American Artificial Intelligence Initiative in 2019 with the mission to promote its leadership in AI research, development, and application. One of the eight national strategies identified in this initiative is to "provide education and training opportunities to prepare the American workforce for the new era of AI" (National Science & Technology Council, 2019).

While computer science and engineering schools have long been offering AI courses, students from other disciplines, like medicine and business, also hope to remain relevant in the age of AI. Equipping themselves with highly sought-after AI knowledge and skills as a

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part of their degree program can help them enhance their chance of employment now and in the future (Stine, Trumbore, & Woll, 2019). As AI technologies become more ubiquitous in the workplace, business and other professionals must become familiar with the principles on which these technologies are built in order to assess the potential they hold along with the risks they carry. To train an AI-ready workforce and prepare students for the era of AI, educators have advocated to incorporate AI training in higher education curricula and learning environments. However, a survey of business school deans shows that many business and management schools are struggling to incorporate AI training into business education (Stine et al., 2019). Our examination of the US News' top 25 graduate business schools of 2021<sup>1</sup> indicates that only 11 universities have offered AI courses in their curricula of business programs (including MBA and Master of Science programs).

One of the challenges that business schools face, when considering offering AI related courses, is the task of *curriculum development* (Stine et al., 2019) due to the lack of pedagogical resources such as model curricula and textbooks. Our review of AI-related textbooks offered by major textbook publishers (e.g., Pearson, McGraw-Hill) shows that most of them are written for technical audiences and those few that target business audiences typically present only a high-level view of AI technologies. The lack of an adequate explanation of how the methods actually work precludes students from understanding their inner workings and becoming competent users capable of appreciating the technology's full potential and limitations. In other words, the key challenge is to find materials to present a high-level perspective supported with just enough technical detail for non-technical audiences.

The difficulty of finding teaching materials in the absence of an appropriate-level textbook is exacerbated by the fact that AI is a broad, complex discipline which encompasses a wide range of theories, models, and techniques. For example, the most popular AI textbook (Russell & Norvig, 2020) consists of 28 chapters, each of which covers dozens of concepts, multiple algorithms and methods; and it is not straightforward to determine what content will serve business students most effectively. Moreover, the field is fast evolving with new algorithms and techniques being rapidly developed, making it a worthy, yet daunting task to keep up with the latest innovations and state-of-the-art technologies. On the other hand, leaving out some of the most foundational concepts in favor of the latest developments will deprive students of a holistic view of intelligence in machines, which may shortchange them in the future.

Our examination of the top 25 U.S. business schools shows that many business programs, while meeting the market demand for business analytics in the past decade, have offered data mining and analytics courses. However, these courses often focus only on introducing statistics and machine learning techniques related to data analysis, and do not adequately prepare students for future work environments where a broad array of diverse AI technologies are used.

Another open question in curriculum development is the selection of effective *teaching methods*. In particular, it remains unclear if programming assignments should be required for AI courses targeting business students. In traditional AI training in computing and engineering disciplines, which stress the development of new methods and applications, programming is used as one of the primary teaching methods. Hands-on practice with AI algorithms and methods is essential to the development of a deeper understanding of the logic and mechanisms of AI techniques as well as their possible biases, limitations, and ethical implications. Without such understanding, future business people and managers may have to treat AI as a black box and leave critical decisions to machines (Burton, Goldsmith, & Koenig, 2017), which is problematic. Although typical teaching methods in management and business courses, such as case study, can be used to present AI concepts and techniques at the conceptual level, it is impossible for students to gain the necessary understanding of the techniques by merely reading stories about how companies and organizations employ AI to help with decision-making and problem solving (Ward 2020). On the other hand, since business graduates will unlikely need to handle all the intricacies of AI algorithms and representations, AI training in business schools may not necessarily need to have programming assignments with comparable requirements as in computing and engineering disciplines.

Equally important is the assessment of student *learning outcomes* and student *perceptions*, which can be used as an important input for curriculum design and improvement (Allan, 1996). However, since AI courses have not been widely offered in business schools, there has been little research reporting student learning outcomes and their opinions toward various aspects of the curriculum, as well as their perceptions of the relevance of AI education to their future careers.

In this paper we report on experience and outcomes of a graduate-level, introductory AI course at a business university in the northeastern United States. We present curricular decisions and findings from student feedback surveys regarding the overall design and various components of the course. Through this exploratory study, we intend to address several research questions: What *content topics* should be covered in an introductory AI course in a business curriculum? How do students perceive the *effectiveness of the teaching methods*? How do students perceive the *relevance of the course* to their future careers? In addition, we also hope to find out if various factors (e.g., student background) have any impacts on students' *learning outcomes*.

The remainder of the paper is structured as follows. Section 2 reviews the literature on AI education and learning outcomes. Section 3 introduces the course design, discussing the topics covered, teaching methods, and assessment of learning outcomes. The next section presents the analysis and results from the student feedback survey. Section 6 discusses the findings, reflections, and recommendations of designing AI curricula for business education. The last section concludes the paper.

<sup>1</sup> <https://www.usnews.com/best-graduate-schools/top-business-schools/mba-rankings>.

## 2. Literature review

### 2.1. AI education

The academic subject of artificial intelligence is typically found in the curricula of computer science and other engineering disciplines. The Association for Computing Machinery (ACM) and the Institute of Electrical and Electronics Engineers (IEEE) have named artificial intelligence (and intelligent systems) as one of the key knowledge areas of computer science programs (ACM, 2013). Other disciplines, such as information systems (Topi, Karsten, & Brown, 2017; Romney, Guymon, & Romney, 2019), have also been actively promoting AI education. Even some non-technical disciplines have realized the potential disruptive power of AI technologies and their pervasive impacts in the future workplace, and are joining the discussion of AI education. For example, many medical and healthcare academics and professionals believe that the use of AI technologies will be increasingly embedded in medicine (Weisberg & Fishman, 2020), and have called to provide AI training to medical school students (Sapci & Sapci, 2020). Medical areas where AI technologies have already been adopted (e.g., radiology) have experienced a pressing demand from students for AI training (Weisberg & Fishman, 2020). In addition to the higher education sector, K-12 educators are proposing experimental courses and programs to foster AI literacy and competence of young children, helping them get prepared for the AI-rich environments in the future (Brummelen et al., 2020; Eaton et al., 2018; Chiu & Chai, 2020).

Despite the growing enthusiasm for AI education, there is a consensus that introductory AI courses are generally difficult to teach even in engineering programs (Eaton et al., 2018; Harris & Kiefer, 2004; Langley, 2019). This is because the field of AI is broad and complex by nature, encompassing many advanced topics and techniques. In addition, the recent boom in AI research and application is rapidly updating the body of knowledge related to AI, making the subject even more difficult to teach.

One of the challenging tasks of teaching AI is the design of curriculum (Harris & Kiefer, 2004; Langley, 2019). The education literature on AI offers limited insights and guidance. A large amount of recent research focuses on the adoption and use of AI as an educational and pedagogical technology to enhance student learning experience (Buckingham & Luckin, 2019). These technologies include personalized tutoring systems, educational robots, and adaptive learning environment (Chassignol et al., 2018; Goel & Yoyner, 2017; Rad, Roopaei, & Beebe, 2018). Pedagogical research on the design of curriculum and teaching methods for AI training is relatively rare.

In the practice of AI education, the annual symposium on Educational Advances in Artificial Intelligence (EAAI) offers a venue for educators to discuss pedagogical issues and share resources related to teaching and using AI in higher education.<sup>2</sup> The popular "AIMA" textbook, *Artificial Intelligence: A Modern Approach* (Russell & Norvig, 2020), has been widely adopted by more than 1500 schools in 135 countries and regions (AIMA, 2021). This book employs a holistic, comprehensive approach and presents a spectrum of AI topics ranging from AI history and foundations, logic and knowledge representation, search and planning, machine learning and natural language processing, to robotics. Although a curriculum can be designed based on this book, it is nearly impossible for any single-semester AI course to teach all topics in this 1132-page book at depth. Hence, most Introductory AI courses in computer science and engineering programs select only a subset of topics (Langley, 2019), and the curricula of different institutions and programs are rather diverse (Harris & Kiefer, 2004).

In terms of teaching methods, the traditional lecture-based method still dominates the pedagogy of AI in engineering programs. Supporting methods such as in-class exercises, discussions, labs, and projects are also frequently used. Specifically, computer programming assignments, which require students to implement AI algorithms and/or develop AI systems and applications, are often used as a critical learning tool to foster and strengthen students' AI knowledge and skills (Langley, 2019).

### 2.2. AI in business curricula

The literature on teaching AI related topics in business curricula is very limited and most prior research is outdated. For example, a few papers in the 1990s promote applied expert systems in business courses (Liebowitz, 1992), or suggest to integrate AI into the accounting curriculum to help student adapt to working environments where AI technologies are used for auditing, financial and management accounting, and taxation (Baldwin-Morgan, 1995). A recent paper suggests that both undergraduate and graduate accounting curricula be updated to provide students with necessary skills and competences in AI (Ucoglu, 2020).

AI education in business and management schools faces more challenges than engineering schools do (Stine et al., 2019). A significant challenge is the scarcity of pedagogical resources. Design principles and guidelines for curriculum development are absent from the literature; and disciplinary associations (e.g., the Association to Advance Collegiate Schools of Business - AACSB) have not recommended any model curriculum, although they have anticipated possible impacts of AI on business education (AACSB Business Education Intelligence, 2017) and called to embrace the AI-integrated curriculum (Whitwell, 2020). Most existing pedagogical resources are intended for engineering audiences. For example, the widely adopted AIMA textbook (Russell & Norvig, 2010) contains a large amount of mathematical symbols, formulas, derivations, and computer algorithms. Business students, who do not necessarily have a strong technical background, may find it very difficult, and even intimidating, to read the book. Also lacking in this book is demonstrations and Case studies about how these technologies are used in real-world applications to solve business problems. In addition, some traditional techniques (e.g., expert systems) are rarely used in today's AI applications and many new technologies (e.g.,

<sup>2</sup> <https://aaai.org/Conferences/AAAI-21/eaai-21/>.

convolutional neural networks) are not discussed adequately in this book. Consequently, the contents of this book may appear less relevant from the perspective of business students, who wish to learn the state-of-the-art AI techniques and their applications in organizations. Although some newer books do talk about AI for business (e.g. (Burgess, 2018; Davenport, 2018; Marr & Ward, 2019)), most of these books merely introduce AI techniques at the conceptual level while presenting case studies on the applications of AI in business. Their technical contents lack depth and students may not fully understand the mechanisms, limitations, and more importantly, the possible ethical implications of using these AI technologies in real-world applications (Burton et al., 2017; Eaton et al., 2018).

Moreover, there has not been research investigating whether computer programming, which is a crucial component in AI training in engineering programs, is an appropriate teaching and learning method for business students.

It is worth mentioning that, in recent years, a large number of Massive Open Online Courses (MOOC) programs have emerged featuring AI courses and many AI related tutorials have become available on online platforms (e.g., YouTube). The popular MOOC website, Coursera, offers a series of AI courses. However, the curriculum contents of these online courses and tutorials often are limited to one or more specific topics (e.g., deep learning), without presenting a comprehensive picture of AI. Also, these courses are generally not designed for business students.

### 2.3. Learning outcomes

The assessment of learning outcomes and student perceptions may offer valuable feedback to the design of a curriculum and reveal whether various learning objectives have been accomplished (Allan, 1996). Such analysis is especially important for the identification of room for improvement for AI training in business curricula, a new educational phenomenon of which little findings and insights can be found in the literature. Specifically, the assessment of learning outcomes “makes possible a refined understanding of the learning process and thus permits a drawing of relatively precise implications for the design of instruction” (Gagne, 1974: 51).

Learning outcomes can be assessed in three dimensions: cognitive, skill-based, and affective (Kraiger et al., 1993). Cognitive outcomes refer to the quantity and quality of knowledge and intellectual development through learning. Skill-based outcomes concern the development of technical skills. The affective dimension concerns the attitudes (e.g., satisfaction) toward the learning process. The assessment of cognitive and skill-based outcomes often is based on actual performance measures (e.g., course grades) or student self-reported, perceived (or expected) performance. For example, a meta-analysis of research on high school student performance shows that GPAs, teacher direct assessment, or national achievement test scores (e.g., SAT) are frequently used as indicators of learning outcomes (Taylor et al., 2014). With the absence of actual performance indicators, research has relied on surveys to gather students' perceived performance. For instance, the learning outcomes of an ERP course are assessed based on students' expected grades from their group projects (Alshare & Lane, 2011). In an entrepreneurship program, Hytti et al. (2010) used students' evaluation of the quality and quantity of business ideas as a proxy for their learning outcomes. Lent et al. (1986) used perceived career options, in addition to grades, to measure student performance.

A large body of literature has reported that student learning outcomes may be affected by a number of factors, including course structure and design (Watson et al., 2016; Eom & Shill, 2016; Alshare & Lane, 2011), student self-efficacy (Eom, 2010; Shen et al., 2013; Lent et al., 1986), and instructor knowledge (Alshare & Lane, 2011), etc. Based on the self-determination theory, many studies have also found that both intrinsic or extrinsic motivations of students play an important role in achieving their learning outcomes (Chen & Jang, 2010; Giesbers et al., 2013; Halliwell, Freeman, & Win, 2016; Niemiec & Ryan, 2009; Taylor et al., 2014).

## 3. Curriculum design

In this section, we present the curriculum content, teaching methods, and learning outcome assessment in an introductory AI course offered at a northeastern business university in the United States. This course is a three-credit, graduate-level course designed for a regular 15-week semester. The class meets for 160 min once a week. All master students in business majors, including Master of Business Administration (MBA), Master of Science (MS) in information technology (MSIT), business analytics (MSBA), finance (MSF), and accounting (MSA), can take this course as an elective to fulfill their degree requirements.

### 3.1. Course prerequisites

In order to provide students with an opportunity to experience developing AI-based solutions first-hand, we made a decision to include programming assignments that use Python and its machine-learning libraries. Hence, we require an introductory Python programming course offered to all graduate business students as a prerequisite. Students with equivalent Python programming experience from other institutions or programs can also take this course. In addition, students are expected to have basic knowledge of statistics and linear regression, but are not required to have any technical background.

### 3.2. Content topics

Because of the lack of AI textbooks for business students, this course does not have a required textbook. Two books, *Artificial Intelligence: A Modern Approach*<sup>3</sup> (Russell & Norvig, 2010) and *Deep Learning* (Goodfellow et al., 2016), are used only as references for interested students to find extended reading materials. The majority of the course content is contained in the *lecture notes* and *presentation slides*, which are compiled and developed using materials and resources from various sources including AI books and magazines, research articles, industry studies, business reports, and websites.

The content is divided into several modules, each of which represents an essential subarea of AI, forming a comprehensive picture of AI. These modules include AI foundations and intelligent agents, knowledge representation and probabilistic reasoning, problem solving, machine learning, and ethics. We believe these content topics are important for students to understand the fundamental concepts and principles, and state-of-the-art techniques of AI. Table 1 presents the course schedule with a summary of topics and assignments.

The course starts by introducing the fundamental concepts of intelligent behavior and rational agents, knowledge representation and reasoning, and problem solving as search, using examples from robotics and other domains.

In five weeks, we move to *machine learning*, which is the largest module in this course due to its predominant role in the latest AI technology boom. In this module, we focus on learning agents which are capable of improving their future performance by extracting patterns and relationships from training examples. Two categories of supervised learning techniques are selected: decision trees and neural networks. These two types of techniques have been employed most widely in AI applications.

Decision-tree-based classification approaches have the advantage of being intuitive and easy to interpret. In addition to the classical algorithms, such as ID3, and ensemble methods, such as random forest, we also teach XGBoost (Chen & Guestrin, 2016), the latest development in decision-tree-based approaches. This algorithm has shown outstanding performance compared to other classification algorithms and has become the *de facto* method in many data analytics contests. In this module, students also learn how to evaluate and compare the performance of different classifiers using a set of metrics (e.g., accuracy, sensitivity and specificity, precision, recall, and F-score) and how to interpret the results in the particular context of a business problem.

Neural networks with many hidden layers, a.k.a. deep learning, have been increasingly used in various applications such as speech recognition, language translation, and computer vision. Deep neural networks often have rather complex structures and require a significant amount of parameter tuning and regularization (e.g., L1 and L2 regularizations and dropout layers) to achieve good performance. It is difficult for a user to effectively leverage deep learning before understanding the mechanisms of neural networks. The deep learning part of this module starts from the simplest neural network architecture - a perceptron with only one layer and one node - and progressively moves to more complex architectures. Advanced strategies for parameter tuning and regularization from the recent deep learning literature are also presented. With the basic understanding of deep learning, students get to explore the topic of convolutional neural networks (CNNs), which is the enabling technique for computer vision applications, such as facial recognition, object detection, and autonomous cars. Students learn the fundamental characteristics of CNNs, as well as the recent developments and successful CNN architectures such as VGG (Liu & Deng, 2015) and ResNet (He, Zhang, & Ren, 2016).

This module was originally intended to include recurrent neural networks (RNNs) and basics of natural language processing (NLP). However, the COVID-19 pandemic caused disruptions to the teaching schedule and, as a result, we did not have time to cover RNNs and NLP.

*Ethics* is an essential component of the business-school course on AI. Discussions in class start by recognizing that AI is not omnipotent - it has constraints, limitations, and weaknesses. Having experienced these limitations first-hand through working on their hands-on assignments (for example, it is impossible to interpret the thousands of weights in deep learning and difficult to explain how a neural network reaches a specific conclusion for a classification problem) students appreciate the ethical considerations even more. More broadly, they can see how the adoption of advanced AI technologies may cause unprecedented risks to privacy, social equality, employment, public safety, and global sustainability (Goralski & Tan, 2020). When discussing the ethical issues and possible threats of AI to society and humanity in this module, we encourage students to keep these questions in mind when they use AI technologies in their future work.

### 3.3. Teaching methods

Several teaching methods are used in this course, including lectures, demonstrations, Case studies, in-class exercises, programming assignments, and a term project.

The majority of the class time is devoted to *lectures*. Our self-developed lecture notes introduce and define fundamental concepts, explain important principles, and illustrate key algorithms and mechanisms using a large number of examples. Small *in-class exercises* are given from time to time to help students understand and practice what they have learned.

Python programs that implement various algorithms (e.g., naive Bayes classifier and breadth-first search) are presented to demonstrate how to transform abstract algorithms to working computer artifacts that can be used for data analysis, problem solving, and other applications. Several widely adopted data analysis and machine learning libraries, including numpy, pandas, scikit-learn, tensorflow, and keras, are introduced in these demonstrations.

<sup>3</sup> We used the third edition of this book (Russell and Norvig 2010) in 2020 when the 4th edition (Russell and Norvig 2020) was not yet published.



Case studies on AI technologies in business are used either to motivate a topic or to connect what students learn in the classroom to real-world applications. For example, students study and discuss how a Bayesian belief network can be used to analyze customer survey data and identify factors affecting customer satisfaction in Queensland Rail in Australia (Chakraborty et al., 2016); how ontologies and semantic networks could enable an intelligent agent to acquire “common sense” about the world; and how IBM’s Watson helps companies automate customer services (Nuruzzaman & Hussain, 2019).

Students are given five *homework assignments* in this course. These assignments are designed to help students grasp the course material. Through programming, students are able to focus on important details and acquire a deeper understanding of the AI techniques they study. For example, by implementing the feed forward and back propagation, as well as the gradient descent algorithms, students get to comprehend how this process is carried out inside of a neural network. Some of the assignments leverage popular Python libraries (e.g., scikit-learn and keras). With the hands-on experience using these libraries obtained in this course, students will be able to develop machine learning and data analytics solutions in their future work.

To overcome the lack of strong programming skills, we prepare the students for each programming assignment by providing sample programs, which they can study and use as a starting point. Students then implement additional functions and modules to fulfill the assignment requirements. For example, for the fourth assignment (multilayer neural networks), students are given a sample program that implements a perceptron with only a single layer. Students extend this program by implementing forward and backward propagation through multiple layers.

For the *term projects*, students form groups of 2–3 members. Early in the semester, each group must submit a proposal for their term project, describing a real-world problem that they want to address using AI technology. Throughout the semester, they work on data collection and preparation, algorithm selection and implementation, and term paper. At the end of the semester, they present their results and findings.

Except for the individual homework assignments, students are encouraged to participate in communication and interactions in and outside the classroom, such as in-class exercises and discussion, office hour visits, and group project collaboration.

**Table 1**  
Course schedule, content topics, and homework assignments.

| Week | Topics                                                                 | Summary                                                                                                                                                                                                                                          | Assignments                                                                                                                    |
|------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | AI foundations and intelligent agents                                  | Intelligent agents and rational decision-making; history of AI including the two “AI winters” (Russell & Norvig, 2010) and the resulting paradigm shift from expert systems to machine learning agents; connections of AI to various disciplines |                                                                                                                                |
| 2    | Knowledge representation                                               | Representing intelligent agent’s internal state and the state of environment using propositional and first-order logic; partially observable and nondeterministic environments                                                                   |                                                                                                                                |
| 3    | Probabilistic reasoning                                                | Probabilistic reasoning for dealing with uncertainty; uncertainty modeling using Bayesian belief network; domain knowledge modeling through ontology and semantic networks                                                                       | #1: Knowledge representation: written problems on first-order logic and probabilistic reasoning                                |
| 4    | Problem solving: search                                                | Strategies for agent problem-solving as search; classical search algorithms: breadth first search (BFS), depth first search (DFS), uniform cost search, and A <sup>a</sup>                                                                       |                                                                                                                                |
| 5    | Problem solving: planning                                              | Planning problem representation and solution as state-space (BFS, DFS, Graphplan) and plan-space search (UCPOP)                                                                                                                                  | #2: Search algorithms: Uniform cost search and A <sup>a</sup> (Python)                                                         |
| 6    | Machine learning: decision-tree-based classification                   | Intelligent agents capable of improving their performance through learning from examples; supervised and unsupervised learning; Decision tree algorithms (ID3)                                                                                   |                                                                                                                                |
| 7    | Machine learning: Ensembles                                            | Ensemble methods: Random forest and XGBoost (Chen & Guestrin, 2016); classifier performance metrics (accuracy, sensitivity and specificity, precision, recall, and F-score)                                                                      | #3: Decision tree & ensembles: classification of bank data using decision tree, random forest, and XGBoost algorithms (Python) |
| 8    | Midterm exam                                                           |                                                                                                                                                                                                                                                  |                                                                                                                                |
| 9    | Deep learning: basics of neural networks                               | Fundamental characteristics and architecture of neural networks; feed forward propagation, backpropagation, gradient descent                                                                                                                     | # 4: DNNs: implementation of multilayer neural networks (Python)                                                               |
| 10   | Deep neural networks (DNNs) and regularizations                        | Advanced strategies for parameter tuning, regularization, and optimization from the recent deep learning literature                                                                                                                              |                                                                                                                                |
| 11   | Convolutional neural networks (CNNs)                                   | CNN architectures; convolution and pooling; recent developments in CNNs and successful architectures: VGG (Liu & Deng, 2015) and ResNet (He et al., 2016)                                                                                        | #5: CNNs: image processing using convolutional neural networks (Python)                                                        |
| 12   | Recurrent neural networks and natural language processing <sup>a</sup> | Recurrent neural networks (RNNs) and basics of natural language processing (NLP)                                                                                                                                                                 |                                                                                                                                |
| 13   | Ethics of AI                                                           | Ethical issues and possible threats to society from using AI; risks of adopting AI technologies to privacy, social equality, employment, and public safety                                                                                       |                                                                                                                                |
| 14   | Final exam                                                             |                                                                                                                                                                                                                                                  |                                                                                                                                |
| 15   | Term project presentation                                              |                                                                                                                                                                                                                                                  |                                                                                                                                |

<sup>a</sup> This topic was not covered due to the schedule interruptions caused by the COVID pandemic.

### 3.4. Assessment of learning outcomes

Student performance is assessed using homework assignments, exams, and the term project. Cognitive learning outcomes are assessed through short-answer and essay questions in the midterm and final exams. The programming assignments, a coding question in the final exam, and the term project are used to evaluate the skill-based learning outcomes.

Students' term projects are graded based on the interestingness of the business problem, the novelty and complexity of the techniques used, the performance of the algorithms, and the quality of the final presentation and term paper. Students are required to write the term paper as a business report with a professional presentation of the executive summary, the business domain and problem, technical design, evaluation results, and findings and conclusions. In addition to assessing the technical skills, this term paper also helps evaluate students' communication and writing skills.

The affective dimension of the learning outcome is assessed through the post-course survey.

## 4. Research method

### 4.1. Survey methodology

To answer our research questions, we used the survey methodology to gather empirical data on students' perceptions regarding the course. This introductory AI course was offered in spring 2020 for the first time. A pre-course survey was administered in the first week of the semester and a post-course survey in the last week before the final exam. The pre-course survey consisted of questions about the student demographics, background, undergraduate major, prior programming experience, and motivations to take the course. The post-course survey included items about students' attitudes and opinions toward the course content (e.g., topics and homework assignments), teaching methods, and learning outcomes. Table A1 and A2 in Appendix provide the details of the items used in these two surveys.

The 16 students who enrolled in this course were invited to take the pre- and post-course surveys. Both surveys were administered online where students' emails were captured by the survey website (Qualtrics) through individualized invitation links sent to students' email accounts. To prevent the social desirability bias (Campbell & Standley, 1963), students were assured that their responses would not be accessed before their grades were posted.

### 4.2. Measures and variables

Sample statistics regarding students' characteristics (e.g., age, gender, background, etc.) and the motivations for taking the course were gathered from the items in the pre-course survey. Other variables, such as student self-efficacy and the amount of time spent each week on the course, were measured using various items in the post-course survey (see Appendix for items measuring the variables).

Student perceptions toward the content topics and teaching methods were measured by a series of seven-point Likert items, ranging from extremely negative (1) to extremely positive (7). Specifically, students were asked to rate the usefulness of individual topics, their relevance to future work, the helpfulness of homework assignments for learning and their difficulty levels, and the effectiveness of teaching methods.

The affective dimension, satisfaction, of learning outcome was measured by a six-question item with each question asking students' degree of agreement with a statement about their satisfaction toward the course, ranging from strongly disagree (1) to strongly agree (5).

The cognitive dimension of the learning outcomes was measured objectively by taking the average of each student's grades of the midterm exam and the question-answering part of the final exam; and the skill-based outcome was calculated by averaging each student's grades in programming assignments, the coding part of the final exam, and the term project.

**Table 2**  
Linear regression on learning outcomes.

|                                   | Cognitive | Skill-based | Satisfaction |
|-----------------------------------|-----------|-------------|--------------|
| Gender (Male)                     | −0.074    | 0.204       | −0.336       |
| Age                               | 2.703     | 2.702       | −0.141       |
| Years of work experience          | −1.439    | −1.525      | −0.149       |
| Major (MSBA)                      | 1.548     | 2.669       | 0.332        |
| Major (MSIT)                      | 0.421     | 1.765       | 0.391        |
| Major (MSF)                       | −0.562    | 1.068       | 0.558        |
| Background (Engineering)          | −0.433    | −0.783      | −0.365       |
| Prior analytics courses           | −1.183    | −0.405      | 0.164        |
| Motivation (Career opportunities) | 0.624     | 0.28        | −0.602       |
| Motivation (Own business)         | 0.251     | 0.02        | 0.187        |
| Motivation (Interest)             | 1.331     | 0.574       | −0.269       |
| Self-efficacy (Ask teacher)       | −0.885    | −0.777      | 0.42         |
| Self-efficacy (On my own)         | 0.331     | 0.348       | 0.261        |
| Hours spent per week              | −1.33     | −0.442      | 0.612        |
| R <sup>2</sup>                    | 0.94      | 0.95        | 0.92         |

## 5. Analysis and results

### 5.1. Sample statistics

Among the 16 subjects, 10 students were male and six were female; and six students were 21–25 years old, six between 26 and 30, and one 31+ years old. The majority (68.8%) of the students were in the business analytics major (MSBA,  $n = 11$ ), and the rest were in information technology (MSIT,  $n = 3$ , % = 18.8), MBA ( $n = 1$ , % = 6.3), and finance (MSF,  $n = 1$ , % = 6.3), respectively. On average, students had 3.9 years of work experience, with the minimum of 0–1 year and the maximum of 10+ years. Four students (25%) had a background in engineering, four in business or management, four in economics, three (18.8%) in finance and accounting, and one (6.3%) in liberal art. Ten students (62.5%) took the prerequisite course on Python programming and six students (37.5%) took an equivalent Python course from other places. In addition, some students also took other technical courses previously, including database management ( $n = 13$ , % = 81.3%), business intelligence and data warehousing ( $n = 2$ , % = 12.5%), and data mining ( $n = 3$ , % = 18.8%).

Regarding the motivations for taking this course, 15 students (93.8%) hoped to increase their career opportunities, 12 (75%) were interested in the subject of AI, six (37.5%) wished to use the knowledge and skills in their future business or partnership; and two (12.5%) took the course because there was no other “good electives” available. In terms of time spent in studying for this course, more than half of the students (56.3%) spent 5–8 hours per week, two students spent less than 4 hours and one student spent more than 16 hours per week. Fig. 1 (a) and (b) present the distributions of student motivation and time spent, respectively.

In the following, we report the findings about our research questions. We have addressed the first research question on curriculum design and content topics in Section 3. In this section, we report our findings regarding the remaining three research questions on student perceptions of teaching method effectiveness, course relevance, and their learning outcomes.

### 5.2. Perceptions of course design

Students had rather positive attitudes toward the usefulness of the topics covered in this course, with an average of 5.91 out of 7.0. Among the topics, deep learning (multilayer neural networks) was rated the highest (6.56), followed by decision trees and ensembles (6.43), convolutional network networks (6.38), and probabilistic reasoning (6.13). Search and planning, basic AI concepts, and knowledge representation were rated relatively low, with an average of 5.72, 5.31, and 5.06, respectively. Fig. 2(a) presents the average ratings of these topics.

Students generally believed that all teaching methods were quite effective with an average of 6.36 (see Fig. 2(b)). Among the teaching methods, lectures and programming assignments were rated as equally important (6.69) for students to understand the course materials. The followers were in-class exercises (6.44), term project (6.19), and Case studies (5.81).

Students found all five homework assignments helpful for learning (5.67). Especially, the four programming assignments were rated higher than the conceptual homework 1 (see Fig. 2(c)), with an average of 5.80. Interestingly, they did not find these assignments too difficult, with an average of 4.44 (see Fig. 2(d)). Even the difficulty levels of algorithm implementations (e.g., assignments 2 and 4) were not rated more than 5.0. This indicates that the programming requirements were appropriate in this course.

### 5.3. Perceptions of course relevance

Student perceptions toward the relevance of topics to their future work was similar to their perceptions of topic uselessness. The average rating was 5.63. The three machine learning topics were rated higher than other topics with decision trees and ensembles being perceived most relevant (6.5), followed by deep learning (6.44), and convolutional neural networks (6.31). Ethics was also believed to be more important (5.81) than the rest of the topics, including probabilistic reasoning (5.5), search and planning (5.13), and basic AI concepts (5.13). Knowledge representation was rated the lowest (4.69). Fig. 3(a) presents the results about student perceptions of topic relevance.

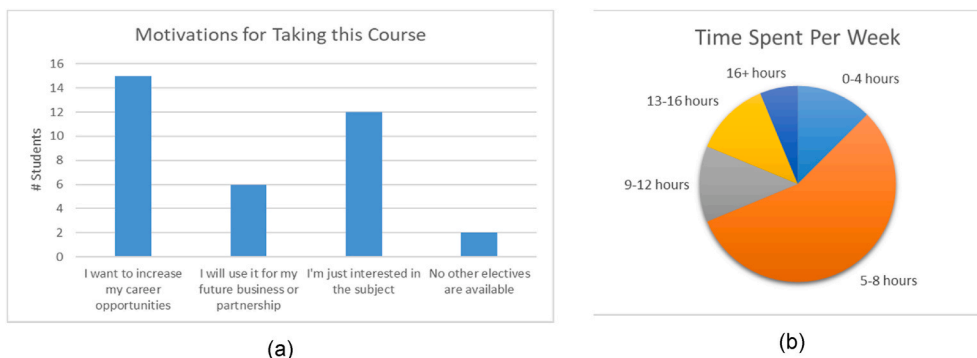
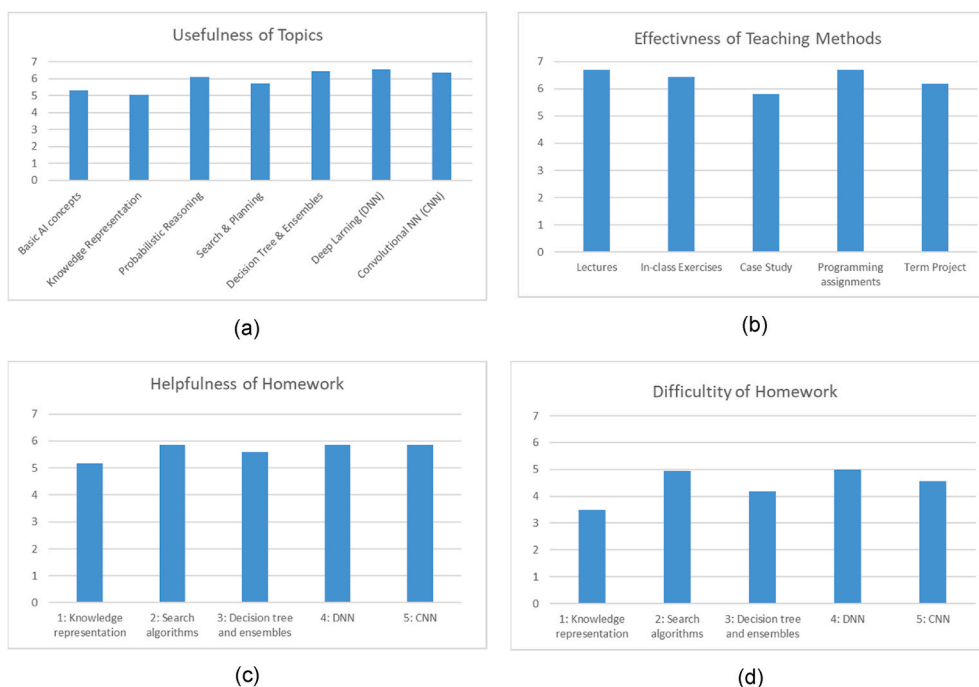


Fig. 1. (a) Student motivations for taking the course. (b) Time spent per week.





**Fig. 2.** Student perceptions toward course design in terms of (a) usefulness of curriculum topics; (b) effectiveness of teaching methods; (c) helpfulness of homework assignments; and (c) difficulty levels of homework assignments.



**Fig. 3.** Student perceptions of (a) the relevance of individual topics to their future work, and (b) expected use of AI knowledge in the future.

Students' responses to the question about their future use of AI knowledge shows that most students ( $n = 15$ ,  $\% = 93.8\%$ ) expected to use AI techniques in data analytics and business intelligence projects (see Fig. 3(b)). Students thought they would also possibly use their knowledge and skills for AI application development ( $n = 2$ ,  $\% = 12.5$ ) or collaboration and communication with AI/IT developers ( $n = 6$ ,  $\% = 37.5$ ).

#### 5.4. Learning outcomes

Students were very satisfied with the course with an average rating of 6.45 (see Fig. 4). Students felt that they had learned fundamental knowledge about AI (6.63), strengthened programming skills (6.25), and learned important problem-solving strategies (6.38). They expected that their AI competence would help them find more job opportunities (6.25) and it would be easier for them to change careers if existing jobs would be automated in the future (6.0). Overall, they felt more confident toward their future careers (6.56).

The small sample size ( $n = 16$ ) prevented us from testing hypotheses on factors' effects on learning outcomes and from drawing statistical conclusions. However, we still hope to explore and analyze our data and get some insights. Table 2 presents the regression analysis of the three dimensions of learning outcomes (affective, cognitive, and skill-based).

Although none of the coefficients (standardized beta) is significant at the 5% level, these results reveal some interesting patterns about the relationships between various factors and learning outcomes. Specifically, male students had better learning outcomes than



Fig. 4. Student satisfaction with the course.

female students in the skill-based dimension but worse in the cognitive and affective dimensions; and an older age helped with the objective learning outcomes but not the affective dimension. Work experience had negative impacts on all dimensions of the learning outcomes. With respect to the MBA major, all majors had positive learning outcomes, except that the finance major performed worse than the MBA major in the cognitive dimension. Students' engineering background and prior analytic courses (e.g., business intelligence and data mining) did not necessarily had positive effects on the leaning outcomes.

Students' intrinsic motivations, compared to the extrinsic motivation ("no other electives were available"), had a positive impact on their cognitive and skill-based outcomes, but not necessarily on their satisfaction. We also found that student's self-efficacy was a predictor of student performance. That is, students who liked to figure out a solution by themselves when encountering a problem tended to perform better than students who chose to discuss with other classmates; and students liked to seek help from the instructor performed worse than students discussing with classmates in objective dimensions of the outcome. In addition, the time spent per week was negatively correlated with the cognitive and skill-based learning outcomes, but positively correlated with satisfaction.

##### 5.5. Qualitative comments by students

Students provided some valuable comments on the course at the end of the post-course survey. Regarding the topic coverage, many students thought that we should have spent less time on knowledge representation and problem-solving, but more time on machine learning. They also wished to learn advanced topics such as recurrent neural networks, natural language processing, robotics, and even reinforcement learning.

Overall, students felt they had learned valuable, useful knowledge and skills. One student noted:

I think I know enough about AI now to be able to go into a client meeting and talk about it and even implement something on a client project.

Some students listed a few techniques that they would be able to use in their work:

I would use Bayesian probability rule, machine learning, and deep learning methods to calculate various analysis in finance. For example, credit scoring (classifier) and predicting future profit/loss (regression, DNN).

Students liked the programming assignments. A student put it nicely:

Homework assignments are very useful to enhance my understanding of AI and improve my programming skills. Especially, assignment 4 gives me a deep impression because it helps me go through the whole process of DNN. Due to this assignment, I understand better about the forward propagation and backward propagation.

A few students had taken or studied MOOC or online certificate courses on machine learning and deep learning. However, they preferred the classroom experience:

I think online courses cannot provide interactions between students and the lecturer especially for some confusing topics such as neural networks. Another thing is that online courses cannot provide us with exercises that are clearly explained and some instructions about how we should build a model from scratch and improve it.

One student who did not have any technical background was really excited about what she had achieved in this course and said:

This class was amazing! Best class I have taken at [the university] and by far the most I've learned in a class. I loved this course!

## 6. Discussion

Artificial intelligence is rapidly reshaping the technological landscape and changing many aspects of human life. Unlike prior transformational technologies that have primarily served as tools, AI and intelligent agents have the potential to become partners of human users and play an important role in decision-making. As educators, we need to prepare students for the increasing pace of technological change brought about by the rapid advancements of AI technologies and their introduction in the workplace (Whitwell, 2020).

So far, AI education has not caught up with the challenge of introducing this complex and very important area of technology to audiences other than students of computing and engineering disciplines. The delivery of AI knowledge to broader audiences, while gaining importance, has not yet been tried nor studied adequately. Although the published literature on AI in education addresses the use of AI to develop novel educational tools, delivery methods and their impact on student learning, very little guidance can be found on actual teaching of AI, especially on the design of the AI curriculum for students with little technical background and skills. The challenge is exacerbated by the lack of teaching materials: textbooks, practice problems, as well as Case studies that are appropriately targeted at the non-technical audience, yet provide sufficient details to develop the understanding of the inner workings of the AI methods, their limitations and the associated risks. Disciplines, such as business, medicine and humanities, lack the experience and expertise in the pedagogy of AI, yet they cannot easily adopt the methods, texts and teaching tools used by the computing and engineering programs due to the obvious differences in the technical readiness of the students. The scarcity of pedagogical resources and studies is not surprising because, the demand for AI competence has risen very recently. However, with the growing presence of AI-based software and automation in workplace, it becomes imperative to start developing effective teaching materials for delivering AI knowledge to non-technical audiences.

In our paper we aim to provide some insights into an approach to AI pedagogy for business schools based on our own experience in designing and delivering a graduate-level course in AI. We hope that our report on the challenges, curriculum design decisions, student feedback and lessons learned from the course provides useful information and a model to business educators wishing to develop AI-related courses.

Several observations summarize our experience:

- *Demand for AI competence motivates students from a variety of business majors.* When we offered this course in spring 2020 for the first time, it was immediately filled up and all 25 seats were taken within a few minutes after the registration system was opened. Several students were placed on a waitlist. Although nine students eventually dropped this course, after learning about the requirements in the first class meeting, 16 students remained enrolled and determined to take the challenge. This implies that there has been a strong demand for AI training by business students. It is unlikely that our university is the only one facing such a demand. Other business schools may consider surveying their students and finding out if there is a potential audience for AI courses.
- *Students with no computing background can successfully acquire AI competence and skills.* We have learned that students do not necessarily have to have a strong technical background, comparable to that of computer science and engineering students, to learn how to use AI methods. Although our students met the prerequisite requirements, which included knowledge of basic Python programming, most students still found mathematical and technical contents, such as differentiation and matrix operations, intimidating. Nonetheless, with the specially tailored introductory materials and examples developed for this course, they were able to comprehend the technical content and achieved satisfactory learning outcomes. Therefore, we believe that business students can successfully acquire AI competence and skills. Furthermore, the hands-on assignments implementing various algorithms and machine learning techniques were invaluable for developing their understanding of the techniques and sense of satisfaction with the course.
- *Course content coverage needs to achieve a balance of fundamentals and state-of-the-art techniques.* AI is a fast-advancing field and new breakthroughs are emerging and are adopted constantly. We recommend a balance between the foundational AI topics (e.g., rational behavior, problem-solving as search, probabilistic reasoning, etc.) and the more recent methods (e.g., deep learning) while designing the curriculum for an introductory AI course. In our experience, the foundational AI topics provided fundamental knowledge for students to get a holistic view of the field and its potential, and to understand the more advanced topics. At the same time students were motivated and interested in learning latest developments in the AI world. It may be quite challenging, however, to achieve a good balance of topics in a single semester. Thus, an alternative approach may be to develop courses with an in-depth focus only on one or two specific topics in AI, for example, robotic automation, or deep learning.
- *Students highly value the practical assignments preferring them to written ones.* A careful design of programming assignments with appropriate level of difficulty can help business students obtain hands-on experience and knowledge and achieve both cognitive and skill-based learning outcomes. Future business people and managers may not need to program to develop AI applications. However, they risk making bad decisions or losing critical control and decisions to machines if they do not have a sound understanding of AI concepts and principles. Completely relying on black-box solutions from off-the-shelf AI software packages will not make them better off, leaving them ignorant of the mechanisms, limitations, and constraints of different AI algorithms. Therefore, hands-on, programming exercises need to be incorporated into the curriculum to help students gain a deeper understanding of the contents, without overwhelming them. Our survey results show that students highly valued the hands-on work and they perceived the assignments and topics that involved programming to be more helpful than the written problems. Likewise, the topics not linked with building a software artifact were considered less important. Hence, we encourage choosing or creating assignments which involve building a working artifact (see <http://modelai.gettysburg.edu/> for a curated assignment repository).

- *Inclusion of ethics of AI is essential and can be benefitted by students' experience in implementing their own AI-based artifacts.* The topic of ethics of AI is worth special attention in a business school, graduates from which will more likely become consumers of the technology and therefore need to be aware of the potential harm and risks associated with using automation. There are published Case studies, such as the one on Microsoft's racist chatbot (Burton et al., 2017), which can be used to stimulate the ethics discussion and motivate students to think critically about various ethical issues and implications of using AI technologies in practice. In our course, in addition to reading about technology exhibiting potential bias due to the properties of the training data, risks and consequences of using imperfectly built tools, the students who had experienced the process of collecting data and developing machine learning programs through course assignments and the term project, gained an informed perspective and an appreciation of the importance of ethical issues.

We would like to point out that what we present in this paper is the design for a standalone AI course. It is also possible to incorporate or integrate AI topics and materials in existing business courses. For example, the introduction of machine learning technique can be integrated into an accounting course to expose students to machine learning supported accounting, auditing, and taxation tools.

The AI Topics website (<https://aitopics.org/>) maintained by the Association for the Advancement of Artificial Intelligence (AAAI) contains a wealth of searchable academic and social media resources, which AI educators may find useful for curriculum development. Given the dearth of materials for teaching AI to non-technical audiences, we call for the creation of web-based portal for sharing teaching materials and experiences of teaching AI to business school students similar to one maintained by the AIMA authors (AIMA, 2021) at <http://aima.cs.berkeley.edu/instructors.html>. We think business disciplinary associations (e.g., the AACSB) can play an important role by offering such a platform for knowledge and resource sharing.

## 7. Concluding remarks

The contribution of our work presented here is threefold: First, this research bridges the gap in the education literature about teaching AI in business schools. We have shown that it is possible and feasible to provide business students with AI training, which is perceived to be highly relevant to students' future careers. Second, by presenting the process and product of our curriculum design, we demonstrate what an introductory AI course may cover in term of contents and how different teaching methods may effectively facilitate students' learning. Third, we offer several practical guidelines for the curriculum design that other business educators may adopt in designing their AI training courses.

Our research has several limitations. First, our study is based on a single course at a single business school. Our findings and observations may not be generalizable to other institutions which have different student populations, background, and expectations. Second, we could not draw any statistical conclusions regarding factors affecting learning outcomes because of the small sample size. As a result, our research is merely a Case study and our findings may have limited applicability.

Our future studies will seek to address the research questions in a larger scale. With continued offerings of this course in our institution and improved curriculum design based on student feedback, we hope we will be able to gather more data and insights into the pedagogy of AI in business schools.

## Appendix

**Table A1**

Variables and items in the pre-course survey.







| Variables     | Items                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Age           | What is your age?                                                                                                                                                                                                                                                                                                                                                                                                            |
| Gender        | What is your gender?<br><input type="radio"/> Male <input type="radio"/> Female                                                                                                                                                                                                                                                                                                                                              |
| Major         | What is your major?<br><input type="radio"/> MSBA <input type="radio"/> MSIT <input type="radio"/> MBA <input type="radio"/> Other, please specify                                                                                                                                                                                                                                                                           |
| Background    | What is your background?<br><input type="radio"/> Business or management <input type="radio"/> Finance <input type="radio"/> Accounting<br><input type="radio"/> Engineering <input type="radio"/> Liberal arts <input type="radio"/> Other, please specify                                                                                                                                                                  |
| Experience    | How many years of work experience do you have?                                                                                                                                                                                                                                                                                                                                                                               |
| Prior courses | Have you taken any of the following courses (check all that apply)?<br><input type="checkbox"/> Python <input type="checkbox"/> Database <input type="checkbox"/> Business Intelligence <input type="checkbox"/> Data Mining                                                                                                                                                                                                 |
| Motivation    | What motivated you to take this course (check all that apply)?<br><input type="checkbox"/> I want to increase my career opportunities<br><input type="checkbox"/> I will use AI knowledge and skills in my future business or partnership<br><input type="checkbox"/> I am just interested in the topic<br><input type="checkbox"/> No other electives are available<br><input type="checkbox"/> Other, please specify _____ |

**Table A2**

Variables and items in the post-course survey.

| Variables                        | Items                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <i>Self-efficacy</i>             | When you were stuck on a problem, how often do you (sum to 100%)<br>- Ask the teacher for help _____%<br>- Figure it out on my own _____%<br>- Ask my classmates _____%                                                                                                                                                                                                                                                                                       |                                                                                                |
| <i>Hours spent per week</i>      | On average, how many hours did you spend outside of class working on assignments, readings, or projects for this course?<br>○ 0-4 hours                      ○ 5-8 hours                      ○ 9-12 hours<br>○ 13-16 hours                      ○ 16+ hours                                                                                                                                                                                                  |                                                                                                |
| <i>Topic useful</i>              | Do you think that the following topics are useful for you to understand AI technologies?<br>- Basic AI concepts<br>- Knowledge representation<br>- Probabilistic reasoning<br>- Search & planning<br>- Decision tree & ensembles<br>- Deep neural networks (DNNs)<br>- Convolutional neural networks (CNNs)                                                                                                                                                   | 7-point Likert scale ranging from extremely useless to extremely useful for each topic         |
| <i>Teaching method effective</i> | How do you rate the effectiveness of the following teaching methods?<br>- Lectures<br>- In-class exercises<br>- Case studies<br>- Programming assignments<br>- Term project                                                                                                                                                                                                                                                                                   | 7-point Likert scale ranging from extremely ineffective to extremely effective for each method |
| <i>Homework helpful</i>          | Do you think that the homework assignments help you understand the course materials?<br>- 1: Knowledge representation<br>- 2: Search algorithms<br>- 3: Decision tree & ensembles<br>- 4: Deep neural networks<br>- 5: Convolutional neural networks                                                                                                                                                                                                          | 7-point Likert scale ranging from extremely unhelpful to extremely helpful for each assignment |
| <i>Homework difficult</i>        | How do you rate the difficulty level of these homework assignments?<br>- 1: Knowledge representation<br>- 2: Search algorithms<br>- 3: Decision tree & ensembles<br>- 4: Deep neural networks<br>- 5: Convolutional neural networks                                                                                                                                                                                                                           | 7-point Likert scale ranging from extremely difficult to extremely easy for each assignment    |
| <i>Topic relevant</i>            | To what extent do you think the topics covered are relevant to your future work?<br>- Basic AI concepts<br>- Knowledge representation<br>- Probabilistic reasoning<br>- Search & planning<br>- Decision tree & ensembles<br>- Deep neural networks (DNNs)<br>- Convolutional neural networks (CNNs)<br>- Ethics of AI                                                                                                                                         | 7-point Likert scale ranging from extremely irrelevant to extremely relevant for each topic    |
| <i>Satisfaction</i>              | To what extent do you agree with the following statements?<br>- I have learned fundamental knowledge about AI<br>- I have strengthened my programming skills<br>- I have learned useful problem-solving strategies<br>- My AI competence will help me find more job opportunities in the market<br>- It will be easier for me to update my knowledge and change careers if existing jobs are automated<br>- I have become more confident for my future career | 7-point Likert scale ranging from strongly disagree to strongly agree                          |
| <i>Future use</i>                | Where do you think you will use your AI knowledge and skills in your future work (check all that apply)?<br><input type="checkbox"/> Data analytics and business intelligence projects<br><input type="checkbox"/> AI application development<br><input type="checkbox"/> Collaboration and communication with IT/AI developers<br><input type="checkbox"/> Other, please specify                                                                             |                                                                                                |

## References

- AACSB Business Education Intelligence. (2017). *Technologies with potential to transform business and business education: Artificial intelligence*. AACSB. [https://www.aacsb.edu/-/media/aacsb/publications/research-reports/ai\\_writeup\\_final.ashx](https://www.aacsb.edu/-/media/aacsb/publications/research-reports/ai_writeup_final.ashx). (Accessed 3 April 2021).
- ACM. (2013). *Computer Science Curricula 2013: Curriculum Guidelines for Undergraduate Degree Programs in Computer Science*. New York, NY: ACM Press.
- AIMA. (2021). *Artificial intelligence: A modern approach*. Available at: (accessed Feb 13) <http://aima.cs.berkeley.edu/>.
- Allan, J. (1996). Learning outcomes in higher education. *Studies in Higher Education*, 21(1), 93.
- Alshare, K. A., & Lane, P. L. (2011). Predicting student-perceived learning outcomes and satisfaction in ERP courses: An empirical investigation. *Communications of the Association for Information Systems*, 28, 571–584.
- Baldwin-Morgan, A. A. (1995). Integrating artificial intelligence into the accounting curriculum. *Accounting Education*, 4(3), 217–229.
- Brummelen, J. V., Heng, T., & Tabunshchik, V. (2020). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum and development tools. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 35 pp. 15655–15663, 17.
- Buckingham, S. J., & Luckin, R. (2019). Learning analytics and AI: Politics, pedagogy and practices. *British Journal of Educational Technology*, 50(6), 2785–2793.
- Bughin, J., Seong, J., Manyika, J., et al. (2018). *Notes from the AI frontier: Modeling the impact of AI on the world economy*. Reportno. Report Number], Date. Place Published]: Institution.
- Burgess, A. (2018). *The Executive Guide to Artificial Intelligence*. Cham, Switzerland: Springer International Publishing AG.
- Burton, E., Goldsmith, J., Koenig, S., et al. (2017). Ethical considerations in artificial intelligence courses. *AI Magazine*, 38(2), 22–34.
- Campbell, D. T., & Standley, J. C. (1963). *Experimental and Quasi-experimental Designs for Research*. Boston, MA: Houghton Mifflin Company.
- Chakraborty, S., Mengersen, K., Fidge, C., et al. (2016). A bayesian network based customer satisfaction model: A tool for management decisions in railway transport. *Decision Analytics*, 3, 4.
- Chassignol, M., Khoroshavin, A., Klimova, A., et al. (2018). Artificial intelligence trends in education: A narrative overview. *Procedia Computer Science*, 136, 16–24.
- Chen, K.-C., & Jang, S.-J. (2010). Motivation in online learning: Testing a model of self-determination theory. *Computers in Human Behavior*, 26, 741–752.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In B. Krishnapuram, M. Shah, A. J. Smola, et al. (Eds.), *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). San Francisco, CA: ACM.
- Chiu, T. K. F., & Chai, C. (2020). Sustainable curriculum planning for artificial intelligence education: A self-determination theory perspective. *Sustainability*, 12(14), 5568.
- Davenport, T. H. (2018). *The AI advantage: How to put the artificial intelligence revolution to work*. Cambridge, MA: The MIT Press.
- Eaton, E., Koenig, S., Schutz, C., et al. (2018). Blue sky ideas in artificial intelligence education from the EAAI 2017 new and future AI educator program. *Special Interest Group on Artificial Intelligence*, 3(4).
- Eom, S. (2010). Relationships among e-learning systems, self-efficacy, and e-learning outcomes: A path analysis model. *Proceedings of the 2010 SIGED: IAII Conference*. Association for Information Systems, Article 23.
- Eom, S. B., & Shill, N. (2016). The determinants of students' perceived learning outcomes and satisfaction in university online education: An update. *Decision Sciences Journal of Innovative Education*, 14(2), 185–214.
- Gagne, R. M. (1974). *Learning for instruction*. Illinois: The Dryden Press Hinsdale.
- Giesbers, B., Rienties, B., Tempelaar, D., et al. (2013). Investigating the relations between motivation, tool use, participation, and performance in an e-learning course using web-videoconferencing. *Computers in Human Behavior*, 29, 285–292.
- Goel, A. K., & Yoynier, D. A. (2017). Using AI to teach AI: Lessons from an online AI class. *AI Magazine*, 38(2), 48–59.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. Cambridge, MA: The MIT Press.
- Goralski, M. A., & Tan, T. K. (2020). Artificial intelligence and sustainable development. *International Journal of Management in Education*, 18(1), Article 100330.
- Halliwell, M., Freeman, M., & Win, K. (2016). Understanding learning outcomes through member needs, motivation and satisfaction: The development of a conceptual model. *Proceedings of the Austra Conference on Information System (ACIS 2016)*, Article 62.
- Harris, H. D., & Kiefer, S. M. (2004). The pedagogy of artificial intelligence: A survey of faculty who teach introductory AI. *Proceedings of the 17th International Florida Artificial Intelligence Research Society Conference*. Menlo Park, CA: AAAI Press.
- He, K., Zhang, X., Ren, S., et al. (2016). Deep residual learning for image recognition. I. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778). San Juan, PR: IEEE.
- Hytti, U., Stenholm, P., Heinonen, J., et al. (2010). Perceived learning outcomes in entrepreneurship education. *Education + Training*, 52(8/9), 587–606.
- Kraiger, K., Ford, J. K., & Salas, E. (1993). Application of cognitive, skill-based, and affective theories of learning outcomes to new methods of training evaluation. *Journal of Applied Psychology Monograph*, 78(2), 311–328.
- Langley, P. (2019). An integrative framework for artificial intelligence education. *Proceedings of the 9th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-19)* (pp. 9670–9677). AAAI Prepress.
- Lent, R. W., Brown, S. D., & Larkin, K. C. (1986). Self-efficacy in the predication of academic performance and perceived career options. *Journal of Counseling Psychology*, 33(3), 265–269.
- Liebowitz, J. (1992). Teaching applied expert systems course: A content outline. *Journal of Information Systems Education*, 4(3), 5–10.
- Liu, S., & Deng, W. (2015). Very deep convolutional neural network based image classification using small training sample size. *Proceedings of the 3rd IAPR Asian Conference on Pattern Recognition (ACPR)* (pp. 730–734). IEEE.
- Marr, B., & Ward, M. (2019). *Artificial Intelligence in Practice*. Wiley.
- Minocha, S., Hristov, D., & Leahy-Harland, S. (2018). Developing a future-ready global workforce: A case study from a leading UK university. *International Journal of Management in Education*, 16(2), 245–255.
- National Science & Technology Council. (2019). *The national artificial intelligence research and development strategic plan: 2019 update*. National Science & Technology Council. <https://www.nitrd.gov/pubs/National-AI-RD-Strategy-2019.pdf>. (Accessed 13 February 2021).
- Niemiec, C. P., & Ryan, R. M. (2009). Autonomy, competence, and relatedness in the classroom: Applying self-determination theory to educational practice. *Theory and Research in Education*, 7(2), 133–144.
- Nuruzzaman, M., & Hussain, O. K. (2019). A survey on chatbot implementation in customer service industry through deep neural networks. *Proceedings of the 15th IEEE International Conference on e-Business Engineering (ICEBE)* (pp. 54–61). IEEE.
- Rad, P., Roopaei, M., Beebe, N., et al. (2018). AI thinking for cloud education platform with personalized learning. *Proceedings of the 51st Hawaii International Conference on System Sciences (HICSS)* (pp. 3–12). Association for Information Systems.
- Ransbotham, S., Kiron, D., Gerbert, P., et al. (2017). Reshaping business with artificial intelligence. *MIT Sloan Management Review*, 59(1), 1–17.
- Romney, G. W., Guymon, J., Romney, M. D., et al. (2019). Curriculum for hands-on artificial intelligence cybersecurity. *Proceedings of the 18th International Conference on Information Technology Based Higher Education and Training (ITHET)* (pp. 1–8). IEEE.
- Russell, S., & Norvig, P. (2010). *Artificial intelligence: A model approach* (3rd). Pearson.
- Russell, S., & Norvig, P. (2020). *Artificial intelligence: A model approach* (4th). Pearson.
- Sapci, A. H., & Sapci, H. A. (2020). Artificial intelligence education and tools for medical and health informatics students: Systematic review. *JMIR Medical Education*, 6(1), Article e19285.
- Shen, D., Cho, M.-H., Tsai, C.-L., et al. (2013). Unpacking online learning experience: Online learning of self-efficacy and learning satisfaction. *Internet and Higher Education*, 19, 10–17.
- Stine, J., Trumbore, A., Woll, T., et al. (2019). *Implications of artificial intelligence on business schools and lifelong learning*. AACSB. <https://www.uniconexed.org/wp-content/uploads/2019/11/AI-Life-long-learning-Final-Report-11.04.19.pdf>. (Accessed 13 February 2021).
- Taylor, G., Jungert, T., Mageau, G. A., et al. (2014). A self-determination theory approach to predicting school achievement over time: The unique role of intrinsic motivation. *Computer Educational Psychology*, 39, 342–358.

- Topi, H., Karsten, H., Brown, S. A., et al. (2017). *MSIS 2016: Global competency model for graduate degree programs in information systems*. ACM Press. <https://www.acm.org/binaries/content/assets/education/msis2016.pdf>. (Accessed 13 February 2021).
- Ucoglu, D. (2020). Effects of artificial intelligence technology on accounting profession and education. *PressAcademia Procedia (PAP)*, 11, 16–21.
- Watson, S. L., Loizzo, J., Watson, W. R., et al. (2016). Instructional design, facilitation, and perceived learning outcomes: An exploratory Case study of a human trafficking MOOC for attitudinal change. *Educational Technology Research & Development*, 64, 1273–1300.
- Weisberg, E. M., & Fishman, E. K. (2020). Developing a curriculum in artificial intelligence for emergency radiology. *Emergency Radiology*, 27, 359–360.
- Whitwell, G. (2020). It's time to embrace the AI-integrated curriculum. *AACSB Insights*. December 15, 2020 <https://bized.aacsb.edu/articles/2020/december/its-time-to-embrace-the-ai-integrated-curriculum>. (Accessed 3 April 2021).